

George Johnson

732-962-2980 | George.johnson022004@gmail.com | Portfolio | GitHub

EDUCATION & INVOLVEMENT

Johns Hopkins University (JHU) | M.S. Applied Economics | GPA: 3.57

Aug 2025 – May 2027

- Focus: Financial Economics | Coursework: Financial Econometrics, Econometrics, Cost-Benefit Analysis, Macro/Micro Theory
- JHU Economics Society – Mentor & Tutor for Econometrics, Game Theory, and Microeconomics

University of Houston (UH) | B.S. Economics & Public Policy | Cum Laude

Aug 2022 – May 2025

- Focus: Quantitative Economics & Statistics | Quantitative Economics Certificate
- UH Economics Society – Member & Tutor; guided lectures & workshops on microeconomics and econometrics
- Code [COOGS] – Peer Mentor; built SQL databases tracking organizational finances

EXPERIENCE

Enbridge – Tidal Energy Marketing | Commercial Analyst (Spring Contract) | Houston, TX

Mar 2026 – Jun 2026

- Audited and reconciled 150+ daily commodity trades across natural gas portfolios, validating risk exposure and mark-to-market P&L
- Implemented Excel-based risk models and SQL databases tracking position valuations, accruals, and deal-capture accuracy
- Synthesized pricing analytics and exposure data into reports for risk committees and commercial stakeholders

Harris County | Financial Analyst Intern | Houston, TX

Jan 2025 – May 2025

- Developed VBA macros automating financial reconciliation and variance detection, reducing processing time by 60%
- Tracked fund allocation and budgetary compliance across 60+ Municipal Utility Districts, flagging anomalies for review
- Oversaw integrity of 100,000+ financial records (1099 forms), ensuring accuracy for year-end reporting and audit readiness

University of Houston | Undergraduate Research Fellow | Houston, TX

Apr 2024 – Apr 2025

- Performed econometric analysis on 1,000,000+ data points using R, Excel, and STATA to test labor-policy hypotheses
- Reviewed 11 academic papers and presented findings on AI's impact on immigrant labor to faculty and peers
- Presented research conclusions to faculty and peers, contributing to discourse on technology-driven labor market disruption

Asia Society | Financial Analyst Intern | Houston, TX

May 2024 – Aug 2024

- Structured cost-benefit models supporting program initiatives that drove 43% revenue growth across the cycle
- Cleaned survey data and refined methodology, lifting survey effectiveness 22% and surfacing untapped demographics
- Delivered financial analyses and data-driven recommendations to leadership, supporting strategic planning for program expansion

TECHNICAL PROJECTS

Cairn — Multi-User Personal Finance & Portfolio Web App | Python, Flask, SQLite, JavaScript (SVG), Argon2id

- Engineered a secure multi-user Flask/SQLite web app with per-user data isolation, Argon2id hashing, and revocable sessions
- Integrated live Yahoo Finance and CoinGecko APIs for real-time quotes; packaged as a one-click Windows build via PyInstaller
- Built an automated trade ledger computing cost basis, realized/unrealized P&L, and allocation across 10 asset classes
- Developed a financial-planning engine modeling cash flow, savings rate, and risk-adjusted 10/20/30-year projections

Crude Oil Price Forecasting Model - Enbridge | Python, OLS Regression, HAC Standard Errors

- Conceptualized three model families (price levels, returns, WCS–WTI differential) for real crude prices, validated walk-forward
- Engineered time-varying geopolitical-risk features with Newey–West HAC errors; differential model hit 71% directional accuracy
- Backtested confidence-threshold strategy yielding ~30pp outperformance vs buy-and-hold at one-eighth the drawdown

PADD 3 Sour & Heavy Crude Supply/Demand Balance Model Gulf of Mexico (GoM) - Enbridge | Excel

- Engineered month-by-month Excel model balancing Gulf Coast sour/heavy supply from GoM, imports, and Canadian pipeline flow
- Sized heavy demand from refinery delayed-coking capacity and reconciled inter-PADD pipeline movements to reported data
- Parsed 10+ years of EIA history into seasonal norms with configurable run scenarios and monthly surplus/deficit outputs

Economic Policy Impact Simulation | MATLAB → Python (NumPy, Pandas)

- Visualized econometric models showing the effects of tax and tariff policy changes on GDP, GDP per capita, and wealth distribution
- Proposed targeted policy reforms to mitigate projected stagflation risks based on model outputs
- Migrated legacy MATLAB models to Python, enabling more flexible analysis, forecasting, and reproducibility

SKILLS

Technical: Python, R, SQL, VBA, STATA, MATLAB, LaTeX, Excel (Macros, Pivot Tables, XLOOKUP), HTML, JSON

Risk & Analytics: Regression Modeling, Causal Inference, Stress Testing, Credit & Counterparty Risk, Variance Analysis, Forecasting

Software: Microsoft Office Suite, Power BI, GitHub, VS Code, Claude Desktop, Google Scholar, Overleaf

Certifications: Quantitative Economics (UH) | Statistics in Python (DataCamp) | Power BI (LinkedIn) | AI Fluency (Anthropic) |

Responsible Conduct of Research (CITI)